

Data and Code Availability in Sports Science: A Registered Report

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Abstract

Sharing scientific raw data and analysis code facilitates cumulative scientific work and enables assessments of reproducibility and robustness. In sports science, preliminary evidence suggests that data and code availability may be low, but previous studies have been based on small cross-sectional samples. To provide a larger audit of the literature, we will document data and code sharing in all original articles published in Q1 sport science journals over the past ten years. We will calculate the percentages of reported as well as actual public availability of data and code and investigate whether general time trends exist. Data and code sharing policies of the analyzed journals will be surveyed to examine associations between policy changes and sharing rates. We provide a resource with all articles in the sample that have shared data and/or code for reuse. The findings of this study can be used to inform future data and code sharing policies in sport science.

Keywords: research transparency; open data; open code; research materials

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28 1 Introduction

29 In light of the reproducibility crisis in psychology and other scientific disciplines and of evidence
30 suggesting that reproducibility may also be low in sports sciences, access to the raw study data is
31 becoming increasingly important to establish robust evidence. Thereby, sharing research data and
32 analysis codes from empirical studies can be beneficial for several reasons. Data and code sharing
33 enables assessments of reproducibility (whether the same analysis on the shared data yields the
34 same results) and analytical robustness (whether reasonable deviations in analytical decisions
35 yield similar results) (e.g., Brodeur et al., 2026). Shared data and code also allow reuse by other
36 researchers, for example, to test alternative theories using the same dataset, combine results
37 across studies (e.g., with individual participant meta-analyses), inform future analyses (e.g., as
38 input parameters for power analyses or priors for Bayesian analyses), or adapt computational
39 workflows to related research problems. From a normative perspective, sharing data and code
40 might also be the right thing to do to advance science, particularly when the research has been
41 publicly funded (Arzberger et al., 2006).

42 Nevertheless, sharing data may also be problematic in some cases. Privacy concerns, particularly
43 in studies of small or vulnerable populations, may prohibit the sharing of data because of the risk
44 of identifying individuals. Proprietary datasets may not be available for sharing because of
45 commercial interests. However, methods for de-identification and simulation of pseudo-datasets
46 have been developed (Naughton et al., 2023; Warmenhoven et al., 2025). With these methods, that
47 arguably require additional effort, data for reuse and reproducibility analyses can be shared most
48 of the time. For analysis code, privacy concerns usually do not apply because analysis code
49 typically does not contain sensitive information about research participants. However, sharing
50 analysis code requires the analysis to be conducted in a programming language, or a code-
51 generating program, which may not always be the case. At the level of the researcher, preparing
52 data and code for sharing may require additional work and training, which some may perceive as
53 unnecessary if it is not incentivized.

54 Over the past two decades, some disciplines like psychology or ecology have seen a moderate
55 increase in data and code sharing (Hardwicke et al., 2024; Tedersoo et al., 2021). In contrast, a
56 meta-analysis of studies investigating data sharing practices in medicine found that, between
57 2014 and 2020, only reported, but not actual data availability had increased (Hamilton et al.,
58 2023). In addition, data availability statements indicating that “data is available upon reasonable
59 request” often mean that data will not be shared on request in practice (Gabelica et al., 2022;
60 Hussey, 2025; Krawczyk & Reuben, 2012; Tedersoo et al., 2021). Data sharing alone does not
61 guarantee that data can be reused (Towse et al., 2021), since it should be shared with appropriate
62 meta-data and licenses (see the FAIR principles, Wilkinson et al., 2016).

63 Policies shape data and code sharing practices. Case studies of journals enforcing mandatory data
64 sharing show increases in public data availability (Hardwicke et al., 2018; Laurinavichyute et al.,
65 2022). Some fields have gone further: For example, all major economics journals require
66 researchers to submit “replication packages” with data and analysis code alongside their articles,
67 and these usually undergo in-house or external reproducibility checks (Brodeur et al., 2024).
68 However, outside selected fields or selected journals, data and code sharing is generally
69 encouraged rather than mandated (Resnik et al., 2019).

70 In sport science, preliminary evidence suggests that data and code availability is low. Twomey et
71 al. (2021) found that, among 300 randomly sampled articles published in 2019 in three leading
72 sport science journals, only two articles shared data (0.7%). Borg et al. (2020) reported a data
73 sharing rate of 1.8% among 283 original articles randomly sampled from Q1 sport science
74 journals in 2019. Schulz et al. (2022) found that none of the 163 clinical trials in sports medicine
75 and orthopedics published in 2019 and 2020 shared their data. Among 195 studies in sports

medicine published between 2014 and 2018, only 2.6% shared at least some data (Evans et al., 2019). Code availability in sport science has been reported at 0% (Borg et al., 2020; Evans et al., 2019).

Previous analyses of data and code availability in sports science have been cross-sectional and limited to small random samples (≤ 300 articles). Moreover, publication practices may have changed over the last decade, as a number of articles in the field have pointed out the benefits of sharing data and/or code (Abt et al., 2022; Borg et al., 2020; Bullock et al., 2022, 2024; Halperin et al., 2018; Memmert, 2025; Sainani & Chamari, 2022; Tenan & Alejo, 2024). At present, a systematic analysis of data and code sharing practices across the field of sports science is missing. To inform decisions on future sharing policies, a description of the status quo, current trends, and indirect evidence of the effectiveness of past and current policies is needed.

The aim of this study is to descriptively present current data and code sharing practices across the full population of original articles published in Q1 sport science journals over the past ten years. We will investigate whether general time trends exist, and will document data and code sharing policies in sports science journals to exploratively analyze their association with sharing practices. Our findings will provide an assessment of the current state of data and code availability in sports science that can inform future discussions about sharing policies.

2 Methods

2.1 Article Search

All articles published in Q1 sports science journals over the past 10 years will be analyzed. We define eligible journals as those that were listed as a Q1 journal in the subject category “Sports Science” in the annual SCImago Journal Rank for at least three years between 2017 and 2025 (SCImago, 2025). Journals that only publish review articles or mainly publish qualitative research (as indicated on the journal website) will be excluded. A total of 31 journals meet these inclusion criteria: *American Journal of Sports Medicine*; *Archives of Physical Medicine and Rehabilitation*; *Arthroscopy - Journal of Arthroscopic and Related Surgery*; *Biology of Sport*; *Bone and Joint Journal*; *British Journal of Sports Medicine*; *Clinical Orthopaedics and Related Research*; *European Journal of Applied Physiology*; *European Journal of Sport Science*; *European Physical Education Review*; *European Sport Management Quarterly*; *International Journal of Sports Physiology and Performance*; *Journal of Athletic Training*; *Journal of Bone and Joint Surgery*; *Journal of Orthopaedic and Sports Physical Therapy*; *Journal of Physiology*; *Journal of Science and Medicine in Sport*; *Journal of Shoulder and Elbow Surgery*; *Journal of Sport Management*; *Journal of Sports Sciences*; *Journal of Strength and Conditioning Research*; *Journal of the International Society of Sports Nutrition*; *Knee Surgery, Sports Traumatology, Arthroscopy*; *Medicine and Science in Sports and Exercise*; *Physical Education and Sport Pedagogy*; *Psychology of Sport and Exercise*; *Scandinavian Journal of Medicine and Science in Sports*; *Science and Medicine in Football*; *Spine*; *Sport Management Review*; *Sports Medicine*. Five journals will be excluded because they only publish review articles (*Exercise and Sport Sciences Reviews*; *Exercise Immunology Review*; *International Review of Sport and Exercise Psychology*) or mainly publish qualitative work (*Qualitative Research in Sport, Exercise and Health*; *Sport, Education and Society*).

Using Web of Science, we will search for all original research articles published in the included journals between 2017 and 2026 with the search string: (SO=(AM J SPORT MED OR ARCH PHYS MED REHAB OR ARTHROSCOPY OR BIOL SPORT OR BONE JOINT J OR BRIT J SPORT MED OR CLIN ORTHOP RELAT R OR EUR J APPL PHYSIOL OR EUR J SPORT SCI OR EUR PHYS EDUC REV OR EUR SPORT MANAG Q OR INT J SPORT PHYSIOL OR J ATHL TRAINING OR J BONE JOINT SURG AM OR J ORTHOP SPORT PHYS OR J PHYSIOL-LONDON OR J SCI MED SPORT OR J SHOULDER ELB SURG OR J SPORT MANAGE OR J SPORT SCI OR J STRENGTH COND RES OR J INT SOC SPORT NUTR OR

KNEE SURG SPORT TR A OR MED SCI SPORT EXER OR PHYS EDUC SPORT PED A OR PSYCHOL
SPORT EXERC OR SCAND J MED SCI SPOR OR SCI MED FOOTBALL OR SPINE OR SPORT MANAG
REV OR SPORTS MED)) AND PY=(2017-2026) AND DT=(Article). The search includes articles
published either in print or online ahead of print during the specified time frame, so it may include
articles with a print publication date in 2027. We focus on original articles because they are the
primary source of scientific raw data. Review articles or editorials typically use aggregated data
or no data at all, and thus will be excluded by the search.

For classifying article type, we rely on the indexing provided by Web of Science. Consequently, the
final sample may include a small number of articles that do not report original quantitative
research because they were misclassified by Web of Science or categorized as original research
but, for example, were a qualitative study. Manual screening of all articles for eligibility is
unfeasible, and we argue that the benefits of assembling a large corpus of studies outweigh the
risk that a small number of ineligible studies will be incorrectly included. For all included articles,
the dates of acceptance and data base indexing (used as a proxy for online publication date) are
retrieved using the Crossref API via the rcrossref package for R (Chamberlain et al., 2026). We will
remove duplicates by automatically identifying duplicated digital object identifiers (DOIs) and
article titles, with each exclusion manually verified.

2.2 Sample Size and Sample Size Justification

The final sample is expected to consist of around 55,000 articles. This estimate is based on a
preliminary search conducted on 14 September 2026 using the search term above. We do not
perform a power- or precision-based sample size estimation because our sample constitutes the
full population of articles published in Q1 sports science journals over the past 10 years.

2.3 Data Extraction

The full-text manuscripts of all eligible studies will be downloaded using the *Zotero* bulk import
function, or by manually downloading the articles from the publishers' websites. Based on the
pilot data, the bulk import is expected to succeed in approximately 25% of cases. Assuming an
average time of 16.5 seconds required per article for manual retrieval of the PDF, as observed in
the pilot, manually retrieving the remaining articles would require around 189 working hours.
Using *Grobid* via the *metacheck* R package (DeBruine et al., 2026; *GROBID*, 2026), we will parse
the PDF files to extract the article texts. An automated algorithm based on the *Grobid* section
detection and regular expression search will be used to flag articles that may contain statements
of data/code availability and data/code provision (see Appendix for the R code). All articles that
are flagged by the automated approach will be manually reviewed to extract the items of interest,
minimizing the rate of false positives.

For each article, we will determine whether a data availability statement exists (yes/no) and
classify the content of the statement (data included in repository, data included in manuscript,
data available upon (reasonable) request, data cannot be made available). We document whether
the raw data is made available for download. Raw data are defined as the individual participant
data that are subjected to the statistical analyses of an article. The same extraction methodology
will be applied for code availability and code provision.

Among the subset of articles that share data, we document whether the FAIR principles (findable,
accessible, interoperable, reusable) (Wilkinson et al., 2016) are followed. We will apply a simple
and liberal operationalization of the FAIR principles to assess compliance. Specifically, we will
define "findable" as the presence of a persistent and unique digital identifier (e.g., DOI) and
associated meta-data (e.g., authors, year, title); "accessible" as data being accessible via an open
protocol (e.g., HTTP) or a documented authorization procedure; "interoperable" as data being
saved in data formats that are typical for the research field and do not require specialized

proprietary software to be read; and “reusable” as being provided with a license that documents the terms and conditions for reuse.

To contextualize code sharing practices, we will also document the (statistical) analysis software used in all included articles. The software used will be identified by regular expression matching against mentions of common analysis software packages (partly informed by the findings of Borg et al., 2020) in the manuscript text and references (see Appendix for R code). Articles that are flagged as using the software *R* will be manually checked because regular expression matching proved difficult for these cases during the pilot.

2.4 Pilot Analysis and Validation Sample

The full text retrieval and data extraction procedures were piloted to estimate the feasibility of article retrieval procedure and to calibrate the automated extraction algorithm using an independent sample of articles. We screened the SCImago Journal Rank 2025 (SCImago, 2025) in the subject category “Sports Science” for journals that were never listed as Q1 in the category during the previous ten years. We then selected the ten journals with the highest 2025 rankings after excluding three journals that primarily publish non-quantitative research (*International Review for the Sociology of Sport*; *Sociology of Sport Journal*; *Journal of the Philosophy of Sports*). We performed an article search using Web of Science on 28 August 2026 based on the inclusion criteria specified in our article. The exact search term was: (SO=(Journal of Human Kinetics OR Strength and Conditioning Journal OR Journal of Sports Science and Medicine OR Physical Therapy in Sport OR Physician and Sportsmedicine OR Sports Biomechanics OR International Journal of Sports Science Coaching OR Journal of Sports Economics OR Research Quarterly for Exercise and Sport OR International Journal of Sports Marketing Sponsorship)) AND PY=(2017-2026) AND DT=(Article).

The search results were exported as .txt files and imported into *R*, where we took a random sample of $n = 200$ articles to constitute the pilot sample. We then attempted to retrieve the articles’ PDFs as outlined in the manuscript and documented the success rate for automated retrieval and the time required for manual retrieval. To validate the automated screening process, we manually extracted all relevant items from the texts. We then conducted the automated screening and compared the results of the two approaches. While reviewing discrepancies between the automated and the manual extraction, we refined the automated extraction procedure and reran it, calibrating the algorithm.

We could retrieve the full texts for $n = 195$ articles¹. The Zotero import function for PDF retrieval worked for 53 (26.5%) articles. The average time required for retrieving a PDF manually was 16.5 seconds. Manual extraction took an average of 52 seconds per study, while automated extraction took 6 seconds per article.

After calibration, data and/or code availability was automatically flagged in 42 cases, of which 29 corresponded to cases identified through manual extraction, 2 identified statements overlooked during manual extraction, and 11 were false positives (11/195, 6%). Note that these false positives would have been identified during the data extraction for the main project, which includes a manual review of each automatically flagged entry. After calibration, no availability statement was overlooked by the automated procedure in the pilot sample.

Regarding the analysis software used, after calibration a total of 43 discrepancies across 40 articles were found. However, after correcting for errors (overlooked uses of software) in the manual extraction, only 17 discrepancies remained; 3 were false negatives (i.e., the automatic

¹ Five articles were recently published in one journal that was included in our university’s subscription system only for articles assigned to an issue, but not for articles published ahead of print. Because this was a pilot study, we excluded these articles without resampling.

extraction overlooked the use of a software), and 14 were false positives (i.e., the automated extraction incorrectly detected the use of a software). Notably, 12 of the 14 false positives were related to the use of the software *R*, because regular expression matching proved difficult in these cases.

To estimate the error rates of the calibrated automated extraction procedures, we will test them against manual extraction in an independent validation set of $n = 500$ articles. The sample size is chosen to yield a 0.02-wide 95% binomial confidence interval for the error rate, assuming a rate of 1%, using the Agresti-Coull method (Agresti & Coull, 1998). These articles will be randomly sampled from the ten highest ranked journals in the SCImago Journal Rank 2025 (SCImago, 2025) for the Subject Category “Sport Science” that are not included in the original study sample. Otherwise, the same inclusion and exclusion criteria will apply. Independence of the validation sample is ensured by excluding all articles from the calibration sample. We will apply the automated algorithm exactly as specified in this document and compare its results with those of the manual extraction.

For the validation, we specify the following minimum acceptable performance criteria. Flagged stated or actual data/code availability must have a false-negative rate below 1%. For flagged software use both the false-positive and the false-negative rate (not flags for *R* use) must be below 3%. If both criteria are met, the algorithm will be applied to the full sample. If either criterion is not met, the respective function will be calibrated and tested in another independent validation sample of the same size using the same methodology. This process will be repeated until the performance criteria are met.

2.5 Journal Policies

All journals included in the sample will be contacted and asked for the current and past editorial guidelines regarding data and code sharing. In addition, we will review current, as well as past (via the Internet Archive, <https://web.archive.org/>) author instructions for all journals. Based on previous work (Hamilton et al., 2020) and the TOP guidelines (Nosek et al., 2015), we will classify data and code sharing policies into four levels: Level 0 (no policy), Level 1 (sharing encouraged and availability statement mandatory), Level 2 (sharing mandatory), Level 3 (FAIR sharing mandatory and/or in-house reproducibility checks).

2.6 Statistical Analysis

The primary outcome is the percentage of studies in sport science that make raw data and/or analysis code available. We will descriptively estimate this proportion by aggregating over the full ten-year sampling period without weighting.

As secondary outcomes, we will descriptively report the percentage of studies that used data availability statements, the reported data availability, as well as the percentage of studies that shared FAIR-aligned data among those that share data. Descriptive results of the analysis software usage across all articles will be presented to contextualize the findings regarding code sharing.

To investigate whether data and code sharing have changed over time, we will fit two logistic regression models of the form $\text{shared} \sim \text{time} + (\text{time} | \text{journal})$, where *shared* is the outcome indicating whether code/data has been shared, *time* is the grand-mean centered continuous indicator of time of acceptance, and *time|journal* is a random effect on the journal level, that allows baseline sharing rates and time trends to vary by journal. We will interpret the fixed main effect of time (odds-ratio) and its 95%-confidence interval, with a 95% confidence interval not including an odds-ratio of 1 as an indicator of the existence of a time effect (see Study Design Template in Tab. 1).

Question	Hypothesis	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
Has the availability of data (H1) and code (H2) in published sports science articles changed over the last ten years?	The availability of data/code has not changed over the last ten years (H1_0/H2_0). The availability of data/code has changed over the last ten years (H1_1/H2_1).	We will investigate the full population of all original articles published in the last ten years (2017-2026) in all sports science journals that were listed as Q1 in at least three years during this period according to the SCImago Journal Rank.	We will run a logistic model of the form: Shared ~ time + (time journal)	We will investigate the full population of Q1 sports science articles for the last ten years	If the 95%-CI for the main effect of time (odds ratio) excludes 1, we will reject the H_0.	n/a

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261 Current journal policies will be presented descriptively. In an exploratory analysis, we will
 262 investigate the associations between changes in data or code sharing policies and the presence of
 263 data/code availability statements and shared data/code. The percentage of articles including
 264 availability statements and data/code sharing will be visually presented as a 100-day moving
 265 average for selected journals, for which the policy history can be fully reconstructed, with markers
 266 indicating changes in guidelines. This approach allows us to contrast associations between
 267 policies and data/code sharing across journals and within journals over time.

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387 Appendix: R Code for Automated Extraction

```
388 #' Flag data availability statements and repository mention
389 #'
390 #' Detects whether an article contains a data or code availability statement
391 #' and whether it mentions a data repository. Availability statements are
392 #' identified either through a dedicated availability section (identified by
393 #' the grobid import) or by searching
394 #' for common phrases (e.g., "data availability", "code availability", or
395 #' "available upon reasonable request"). Repository mentions are detected by
396 #' searching for references to common repositories such as OSF, Zenodo,
397 #' GitHub, Figshare, Dryad, Dataverse, and others.
398 #'
399 #' @param article An object with class "scivrs_paper", usually imported
400 #'   by the metacheck package
401 #'
402 #' @return A data frame with one row and two logical columns:
403 #' \describe{
404 #'   \item{flag_statement}{`TRUE` if the article contains a data/code
405 #'     availability statement or a dedicated availability section.}
406 #'   \item{flag_repo}{`TRUE` if the article mentions a supported data
407 #'     repository.}
408 #' }
409 #'
410 flag_article <- function(article) {
411   # search for availability section in grobid import
412   if (is.null(article$section)) article <- article[[1]] # get list item if
413   needed
414   sec <- any(article$section$section_type == "availability")
415   # search for strings that indicate availability section
416   st <- sfind(
417     article,
418     c("data availability", "code availability",
419       "(on|by|upon)\\s+(reasonable\\s+)?request",
420       "datasets?\\s+(?:is|are)?\\s*(?:not\\s+)?(?:publicly\\s+)?available",
421       "available\\s+from\\s+(?:the\\s+)?(?:corresponding\\s+)?author"
422     ),
423     search_header = TRUE
424   )
425   # search for strings that match data/code repository
426   repo <- sfind(
427     article,
428     c("repositor", "archive", "\\bosf\\b", "osf\\.io",
429       "open science framework", "researchbox", "zenodo", "github",
430       "figshare", "datadryad", "kaggle", "mendeley", "ukbiobank",
431       "ukdataservice", "dataverse", "clinicalstudydatarequest",
432       "ourworldindata")
433   )
434   data.frame(
435     flag_statement = any(c(sec, st)),
```

```

436     flag_repo = repo
437 )
438 }
439
440 #' Detect statistical software mentioned in an article
441 #'
442 #' Searches an article for mentions of commonly used statistical software and
443 #' programming languages. Detection is based on regular expressions and common
444 #' alternative names where appropriate (e.g., "Statistical Package for the
445 #' Social Sciences" for SPSS or "RStudio" for R).
446 #'
447 #' @param article An object with class "scivrs_paper", usually imported
448 #'   by the metacheck package
449 #'
450 #' @return A data frame with one row containing logical columns indicating
451 #'   whether each software package was detected. The returned columns are:
452 #'   `spss`, `stata`, `jamovi`, `jasp`, `r`, `python`, `prism`, `excel`,
453 #'   `sas`, `lisrel`, `matlab`, `statistica`, `winbugs`, `bioestat`,
454 #'   `stan`, `pspp`, and `medcalc`.
455 #'
456 extract_software <- function(article) {
457   data.frame(
458     spss = sfind(article, c("\\bspss", "statistical package for the social
459 sciences", "\\bpasw", "\\bamos")),
460     stata = sfind(article, "\\bstata"),
461     jamovi = sfind(article, "\\bjamovi"),
462     jasp = sfind(article, "\\bjasp"),
463     r = any(
464       sfind(article, c("(?!-)\bR(?:=\\s(?:!\\s*))",
465 "(?:in|software)\\s+R\\b"), ignore.case = FALSE, perl = TRUE),
466       sfind(article, c("a language and environment for statistical
467 computing", "rstudio", "\\br language"))
468     ),
469     python = sfind(article, c("python", "scipy")),
470     prism = sfind(article, c("graph pad", "\\bprism\\b")),
471     excel = sfind(article, "excel\\b"),
472     sas = sfind(article, "\\bsas\\b"),
473     lisrel = sfind(article, "lisrel"),
474     matlab = sfind(article, c("matlab", "mathworks")),
475     statistica = sfind(article, "\\bstatistica\\b"),
476     winbugs = sfind(article, "winbugs"),
477     bioestat = sfind(article, "bioestat"),
478     stan = sfind(article, "\\bstan\\b"),
479     pspp = sfind(article, "\\bpspp"),
480     medcalc = sfind(article, "medcalc")
481   )
482 }
483
484 #' Helper function to find matches using metacheck::text_search()
485 #'
486 #' Searches an article text via metacheck::text_search() and returns

```

```
486 #' if any matches were found
487 #'
488 #' @param article An object with class "scivrs_paper", usually imported
489 #'   by the metacheck package
490 #' @param string Regular expression pattern passed to `pattern` in
491 #'   metacheck::text_search()
492 #' @param ... Additional arguments passed to metacheck::text_search()
493 #'
494 #' @return TRUE or FALSE, whether a match was found.
495 sfind <- function(article, string, ...) {
496   t <- metacheck::text_search(paper = article, pattern = string, ...)
497   nrow(t) >= 1
498 }
499
```